Econometrics II: Home Assignment I

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October 2025

Problem I

Section A

We note that we obtain the following time-series OLS estimator for the uni-variate case when performing SSR minimisation:

$$\hat{\beta}_1 = \frac{\sum_{t=1}^{T} (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^{T} (x_t - \bar{x})^2}$$

Figure 1: Formula of time series $\hat{\beta}_1$ estimator

We will provide the variance of the OLS estimator as a function of residual variance under assumptions TS 1-4, which we will state below.

- TS1 (Linear in parameters): the specifications of the models 1 - 3 are linear univariate models, so one unit change in x leads to β_1 change in variable y. So it is satisfied for all cases.
- TS2 (Strict exogeneity): $E(u_t|X) = 0$ disturbance at period t is unrelated to independent the variable in any time period.
- TS3 (Sample variation in the explanatory variable): For the univariate case, we will use assumption SLR3 instead of MLR3 which states that the sample outcomes for the explanatory variable are all not the same (non-zero variance).
- TS4 (Homoscedasticity): constant variance of the error term that is independent of the regressor in any time period $Var(u_t|X) = \sigma^2$. Necessary for the efficiency of OLS and the simplification of the computation of the variance.
- TS5 (No autocorrelation): no correlation between the error terms given independent variables $Cov(u_s, u_t|X) = 0, \forall s \neq t$. Reduces to 0 the bias term that appears in the variance calculations. [1]

[1] Follows from the variance of sums formula: $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$ if no independence assumed.

For all models, TS2 and TS5 are satisfied due to u_t being i.i.d. with an expected value of zero. TS4 holds since the variance is assumed to be constant.

In such case, we will use the following variance formula:

$$Var(\hat{\beta}_1) = Var\left(\frac{\sum_{t=1}^{T}(x_t-\bar{x})(y_t-\bar{y})}{\sum_{t=1}^{T}(x_t-\bar{x})^2}\right) =$$

$$= Var\left(\beta_1 + \frac{\sum_{t=1}^{T}(x_t-\bar{x})u_t}{\sum(x_t-\bar{x})^2}\right) =$$

$$= Var(\beta_1) + \frac{\cancel{\sum_{t=1}^{T}(x_t-\bar{x})^2}\, Var(u_t)}{\underbrace{\left(\sum_{t=1}^{T}(x_t-\bar{x})^2\right)^{\cancel{2}}}_{SST_x}} = 0 + \frac{\sigma^2}{SST_x}$$

Figure 2: Variance of the OLS estimator under TS.1-TS.5

We also consider the variance of the disturbance:

$$\hat{\sigma} = \frac{SSR}{df} = \frac{\sum_{t=1}^{T}(y_t-\hat{y}_t)^2}{T-K-1}$$

Figure 3: Error variance estimator under TS.1-TS.5

Variance of the OLS estimator for model 1-3:

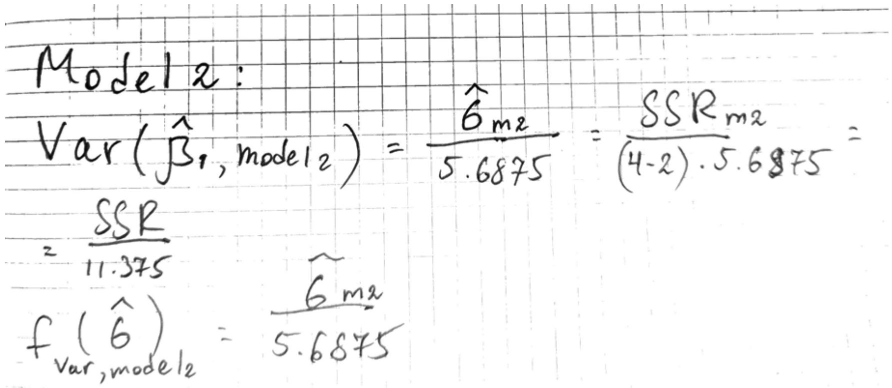
Model 1:

$$Var(\hat{\beta}_{1,model1}) = \frac{\hat{\sigma}_{m1}}{SST_{x,m1}} = \frac{SSR_{m1}}{(T_{m1}-2)\cdot SST_{x,m1}} =$$

$$= \frac{SSR_{m1}}{(31-2)\cdot 30.66855} = \frac{SSR_{m1}}{889.3879}$$

$$f_{Var,model1}(\hat{\sigma}) = \frac{\hat{\sigma}_{m1}}{30.66855}$$

Figure 4: Variance of the 1st model

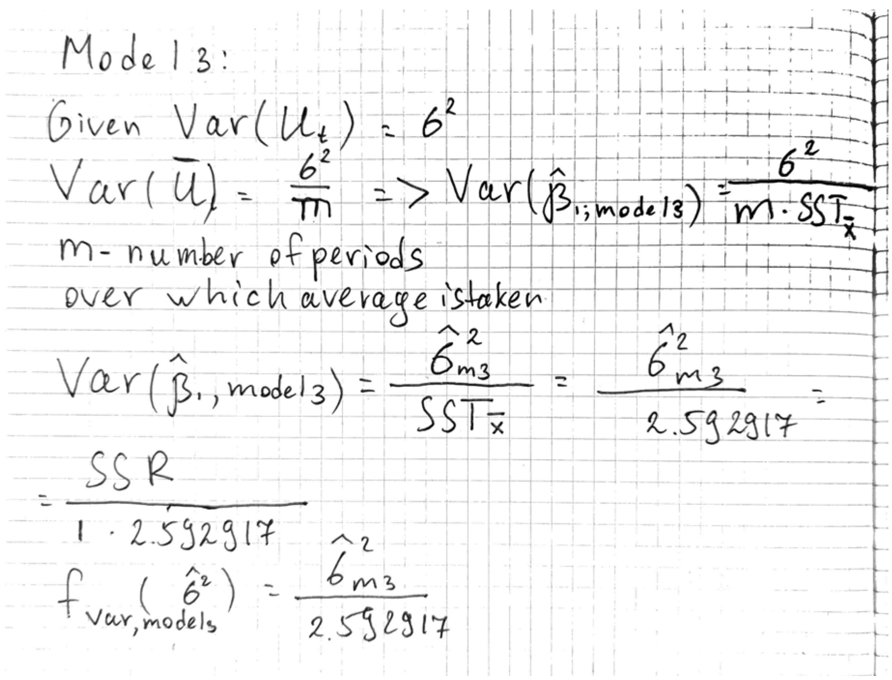


Model 2:

$$Var(\hat{\beta}_1, model_2) = \frac{\hat{\sigma}_{m2}}{5.6875} = \frac{SSR_{m2}}{(4-2)\cdot 5.6875} = \frac{SSR}{11.375}$$

$$f_{Var, model_2}(\hat{\sigma}) = \frac{\tilde{\sigma}_{m2}}{5.6875}$$

Figure 5: Variance of the 2nd model



Model 3:

Given $Var(u_t) = \sigma^2$

$$Var(\bar{u}) = \frac{\sigma^2}{m} \Rightarrow Var(\hat{\beta}_{1}, model3) = \frac{\sigma^2}{m \cdot SST_{\bar{x}}}$$

m- number of periods over which average is taken

$$Var(\hat{\beta}_1, model3) = \frac{\hat{\sigma}^2_{m3}}{SST_{\bar{x}}} = \frac{\hat{\sigma}^2_{m3}}{2.592917} = \frac{SSR}{1 \cdot 2.592917}$$

$$f_{Var, models}(\hat{\sigma}^2) = \frac{\hat{\sigma}^2_{m3}}{2.592917}$$

Figure 6: Variance of the 3rd model

Section B

The variance of the first model 4 should be lower than that of the second. Since it was assumed that both disturbances u_t and u_s are iid with first moment being equal to 0, the estimator of the first variance should be more precise since it has a greater SST_x and more degrees of freedom for the disturbance variance estimation.

In reality, simple linear interpolation can be problematic since it is likely that the data does not behave that way. This will introduce measurement error due to our assumption of an increasing linear trend of the and can lead to violations of strict and even contemporaneous exogeneity. If the trend is decreasing, there is no trend or x grows slower than linear interpolation, then the errors have upward bias. Otherwise, there can be downward bias.

Section C

Assuming that U_t are also i.i.d. with zero expected value as previously specified and assuming that we will use averages as specified in the task, we also assume interpolation for the x values[2], we obtain that variance of the first is lower than that of the third model. This result holds even accounting for the variance of the mean of disturbances, which leads for the estimated variance of disturbance being 10 times lower (see 6). We get this result as we lose more degrees of freedom and get lower $SST_{\bar{x}}$ because of the mean of means.

Similar to the first model, there is also an issue with the specification of the third model. Even though, we get a lower variance of disturbance and values that are less prone to noise due to the means, we might obscure relationships within those averaged periods, e.g. a variable having lagged effect. This might introduce omitted variable bias and lead to a violation of the exogeneity assumptions since the effect of the lag will be attributed to the average and its lagging nature will not be captured.

Section D

Overall, in reality, we would use the second model. Despite it having only 4 observations, it does not make overextended assumptions such as assuming an increasing linear trend of specific size as in the first specification, nor does it disregard the possible lagged effects as in the third model. Although, we must admit that the third model can be used depending on the task at hand. For example, in macroeconomic context: modelling GDP quarterly or annually to see the effect of a policy. If the policy was introduced years ago and had a long-lasting effect, annual GDP could be an option. However, if the effect is not that pronounced, quarterly GDP may suffice. On the other hand, we would not use the first model as we believe there should be very strong theoretical justification for interpolating the values between two timestamps.

Problem II

Section A

Standard economic theory and findings suggest [3] that in the Czech Republic, housing prices are a function of the wage rate, unemployment rate, and building lot price. Specifically, building lot price has played an increasing role in determining housing price, while since the 2008 crisis, wage rates have played less of a role. As the data at hand does not directly represent building lot prices, we will instead use building lot supply as a proxy. Moreover, we will include the unemployment rate as calculated by the Eurostat on a quarterly basis to provide an additional unit of observation based on empirical findings. [4]

Section B

Part I

The data appears to be of a time series format. A variety of macroeconomic indicators and gross values for housing supply are reported for one entity, Czech Republic, on a quarterly basis from 2015 to 2024. There is no cross-section dimension.

[2] One of the TAs and the task indicate that we ought to use interpolated values as in the first model.

[3] Kalabiska, R., Hlavacek, M. (2022). Regional Determinants of Housing Prices in the Czech Republic. Finance a Uver: Czech Journal of Economics and Finance, 72(1).

[4] Eurostat. (n.d.). Population and employment — national accounts (ESA 2010): Total employment, domestic concept (Czechia; Q1 2015–Q4 2024) [Data set]. Retrieved October 28, 2025, from Eurostat Data Browser (online data code: namq_10_pe).

Part II

As mentioned in the previous section, we will also include unemployment per capita as it is a statistically significant influence on housing prices in the Czech Republic. To the best of our knowledge, the technical specification for the Housing Price Index as laid out by Eurostat [5] does not account for inflation, we will create an adjusted index that aims to reflect real asset prices and not nominal values. This will enable comparability across the cross-section of price determinants.

We will also include data from the Czech National Bank that documents both interest rates and loan volume in million CZK to proxy the demand for housing. [6]

Part III

Analysis the data reveals the following:

Variable	**Min.**	**1st Qu.**	**Median**	**Mean**	**3rd Qu.**	**Max.**
Con. Overall	5,057	8,016	8,560	8,820	9,536	12,826
Con. Family Dwellings	2,578	3,148	3,434	3,484	3,903	5,595
Con. Apartments	1,031	1,838	2,706	2,822	3,505	5,692
Ren. Family Dwellings	307	389.5	431.0	426.9	460.0	574.0
Ren. Apartments	444.0	566.2	654.5	689.0	755.2	1,474.0
Con. N/Dwellings	255.0	328.2	395.0	412.3	446.2	845.0
HPI [2010]	101.5	136.1	175.4	187.4	251.7	275.9
Population	10,488,351	10,557,198	10,633,055	10,661,514	10,722,557	10,903,158
Δ Inflation (Q)	0.1%	2.09%	2.1%	3.473%	3.73%	17.58%
GDP (Nom. M CZK)	1,072,634	1,239,052	1,460,981	1,523,537	1,876,787	2,106,384
House Loan IR	1.023%	2.364%	3.18%	3.222%	3.474%	5.508%
House Loan Volume	15,130	26,579	30,459	33,969	37,777	65,675
Employed per 1000	511.3	523.6	533.1	531.2	537.5	548.1
PPI [2021]	81.6	84.4	93.15	97.52	112.83	121.7

It appears that the dataset is both complete and does not appear to have any extreme values. The somewhat low minimum for change in inflation is explained by the quarterly basis of the observations. Other values, such as the price indices are explained by the periods of measurement.

[5] Eurostat. (n.d.). House price index (HPI) (2021 = 100) — quarterly data (dataset code: PRC_HPI_Q; Czechia; Q1 2015–Q4 2024) [Data set]. Retrieved October 28, 2025, from Eurostat Data Browser.

[6] Czech National Bank (CNB). (n.d.). ARAD time series database: Mortgage loans for house purchase — new business, volumes (households), 2015–2024 [Data set]. Retrieved October 28, 2025, from CNB ARAD.

Czech National Bank (CNB). (n.d.). ARAD time series database: Interest rates on housing loans — new business, 2015–2024 [Data set]. Retrieved October 28, 2025, from CNB ARAD.

Part IV

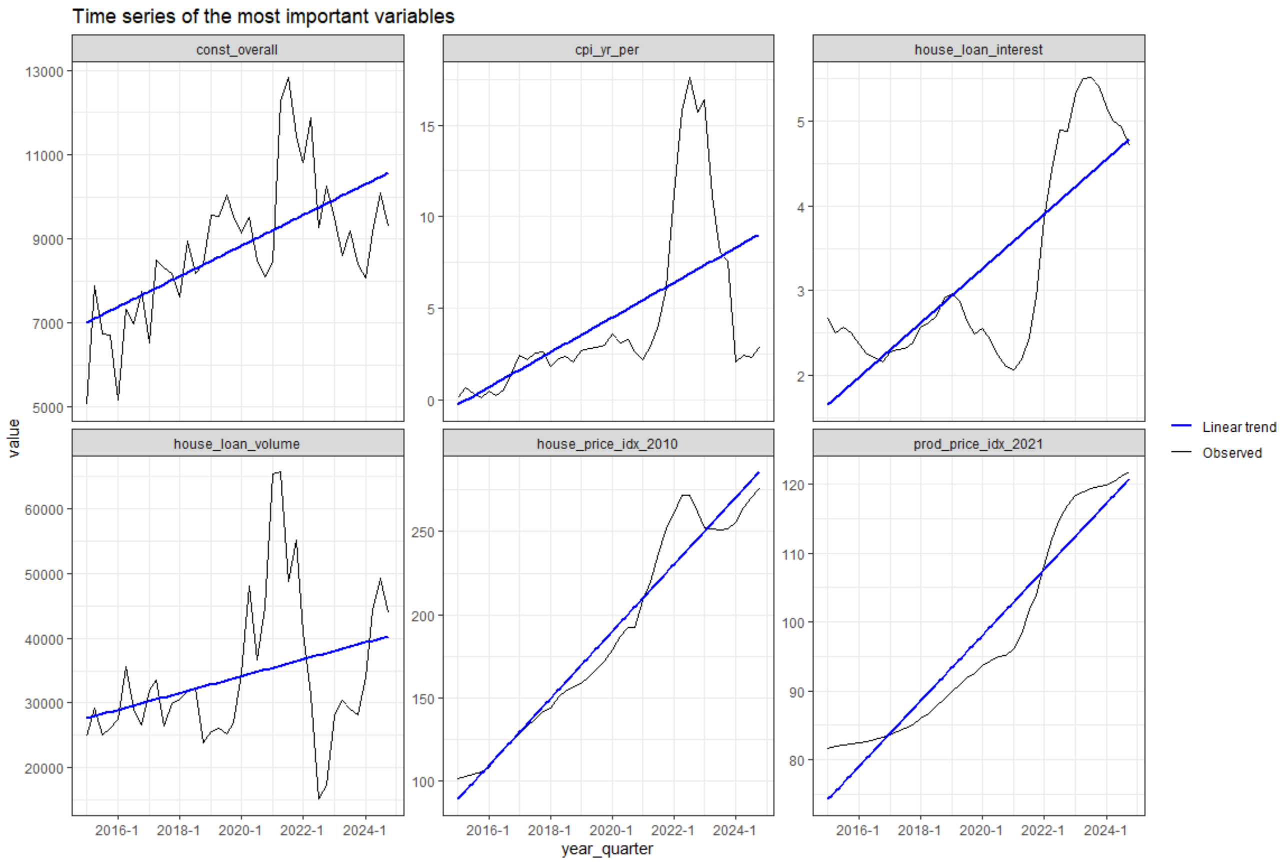


Figure 7: Graphing the most important variables

Part V

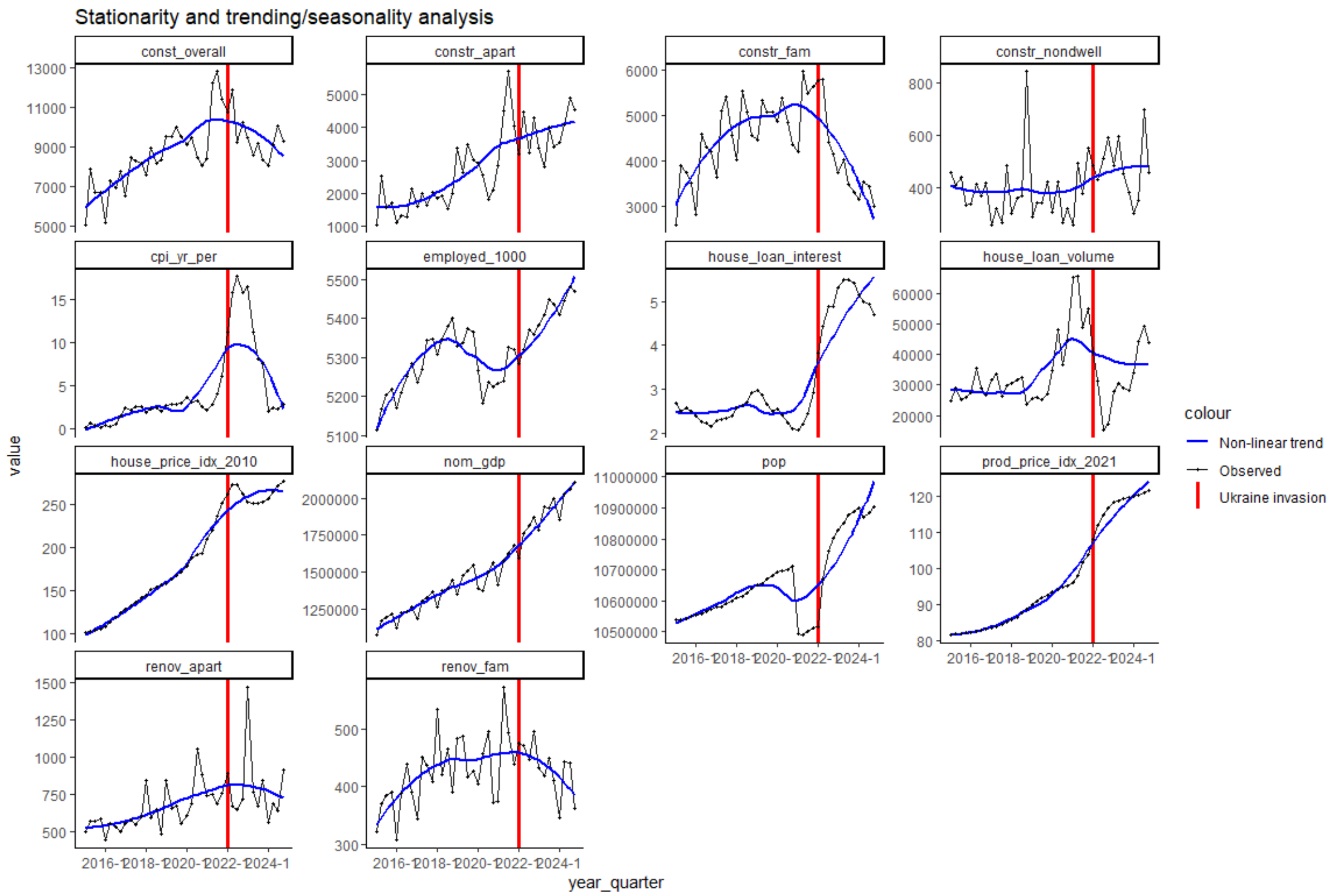


Figure 8: Stationarity and trend seasonality analysis

It appears that variables representing construction and renovation exhibits seasonality, as every first quarter of the year, the incidence of new construction falls. We consider this to be a function of the weather, as most building activities are undertaken during the dry season. In indices tracking price, we note that the covid-era jumps are quite well felt, which indicates another issue of seasonality (even if we have one sample season). Overall, the trends appear to be polynomial rather than linear.

Section C

Part I

The simple linear regression model is as follows:

$HPI_t = \beta_0 + \beta_1 GDP_t + \epsilon_t$

Coefficients	Estimate	Std. Error	t value	Pr(> $\|t\|$)
β_0	-1.44	2.30	-0.63	0.535
β_1	0.00005	0.00002	2.01	0.052 .
Residual standard error: 13.78 on 37 degrees of freedom				
R^2= 0.099, R^2_{adj}= 0.074				
F-statistic: 4.05 on 1 and 37 DF, p-value: 0.0515				

For the simple regression model using OLS, we have a number of problems. First of all, as the graphs in the previous section indicate, we do not have stationary in our data and also have time trends. This means that for different time periods our variables take on different distributions. This also implies heteroskedasticity as our variance will be nonzero. Running a Breusch-Pagan test, we obtain a p-value of 0.7028 which indicates that we cannot reject the null hypothesis of homoskedasticity. Nevertheless, without correction for stationarity, it is quite clear that we have non-constant variance among sample periods.

Because TS1-TS5 are not met, we cannot assume consistency for the OLS estimator. Likewise, because the Gauss-Markov theorem requires TS1-TS5 for estimator efficiency, we cannot say that in this simple model our estimator variance is minimised. Running a Breusch-Godfery test for serial correlation, we obtain a p-value of effectively zero, confirming the presence of serial autocorrelation.

Part II

ADF tests[7] indicate that both variables are not stationary, with high p-values. In order to correct a seasonal trend in nominal GDP and the housing price index, we will apply a first-difference approach for the GDP variable, differencing on a quarterly basis. The corrected model is jointly significant, although it explains a suspiciously large amount of the variation in the model. This is probably due to the close relationship between GDP and the error term.

Coefficients	Estimate	Std. Error	t value	Pr(> $\|t\|$)
β_0	-107.98	18.47	-5.85	9.3e-07 ***
β_1	0.00019	0.00001	16.27	<2e-16 ***
Residual standard error: 21.7 on 38 degrees of freedom				
R^2= 0.875, R^2_{adj}= 0.871				
F-statistic: 265 on 1 and 38 DF, p-value: <2e-16				

Part III

The Durbin-Watson test produced a p value of nearly zero, indicating at least first order autocorrelation. We will employ serially correlated robust standard errors to correct this, accounting for a lag of 4 periods. As the results of the Breusch-Pagan test did not confirm heteroskedasticity among the variances of our variables and we cannot assume strict exogeneity, so we use serial correlation robust standard errors. The results are as follows:

Coefficients	Estimate	Std. Error	t value	Pr(> $\|t\|$)
β_0	-107.98	25.22	-4.28	0.00012 ***
β_1	0.00019	0.00002	10.86	3.2×10^{-13} ***
Residual standard error: 21.7 on 38 degrees of freedom				
R^2= 0.875, R^2_{adj}= 0.871				
F-statistic: 265 on 1 and 38 DF, p-value: $< 2 \times 10^{-16}$				

[7] See the R code

Our ceteris paribus effect is statistically significant and allows us to interpret changes. For example, an increase in GDP by 1 billion CZK would increase the housing price index by approximately 0.19 points, explaining around 87 percent of all variation in the HPI. In the next section, we will expand with the use of several variables.

Section D

Part I

Based on Kalabiska and Hlavacek (2022), we will include employment per capita, construction of dwellings (overall construction minus non-dwellings) and both the interest rate on housing loans as well as the total loan volume. We will also include the producer price index as it can serve as a proxy for the savings of the population, given that with price increases, individuals have less to save and as a result, less housing will be bought. The model is specified as such, with a logarithmic specification for the loan interest rate:

$$HPI_t = \beta_0 + \beta_1 GDP_t + \beta_2 EMP_t + \beta_3 CD_t + \beta_4 ln(IRL_t) + \beta_5 HLV + \epsilon_t$$

The results of our regression are as follows:

	Estimate	Std. Error	t value	$\Pr(> \|t\|)$
β_0	-438.1	217.9	-2.01	0.053
β_1	-0.00005	0.00004	-1.52	0.137
β_2	0.03	0.04	0.72	0.476
β_3	0.0075	0.0013	5.72	0.000002 ***
β_4	-57.7	20.7	-2.80	0.009 **
β_5	-0.00007	0.00023	-0.28	0.781
β_6	5.68	0.85	6.64	0.00000015 ***
Residual standard error: 9.74 on 33 degrees of freedom				
$R^2 = 0.978$, $R^2_{\text{adj}} = 0.974$				
F-statistic: 244.3 on 6 and 33 DF, p-value: < 2.2e-16				

Part II

The Durbin-White test produces a p value close to zero[8], indicating that autocorrelation is present. As discussed in the previous section, seasonality is present. Apart from GDP, we also see that quarterly seasonality is exhibited by construction, employment, and loan volumes, so we will correct by treating for quarterly periods. Therefore, all variables will be corrected for a quarterly period. A Breusch-Pagan test indicates that the variance of the coefficients can have a homoskedastic form. Moreover, the ACF function in R indicated that we a two-period autocorrelation. Therefore, using our serial correlation robust standard errors lagged to two periods and variables corrected for trends, we have the following results:

[8] See R code

Coefficients	**Estimate**	**Std. Error**	**t value**	**Pr(> $\|t\|$)**
β_0	19.49	4.07	4.79	4.3e-05 ***
β_1	-0.00005	0.00005	-1.02	0.315
β_2	-0.114	0.037	-3.04	0.00492 **
β_3	0.00709	0.00162	4.38	0.00013 ***
β_4	60.25	20.19	2.98	0.00560 **
β_5	0.00024	0.00035	0.68	0.502
Residual standard error: 12.7 on 30 degrees of freedom				
R^2= 0.544, R^2_{adj}= 0.467				
F-statistic: 7.14 on 5 and 30 DF, p-value: 1.66×10^{-4}				

Part III

It appears that microeconomic factors explain the housing price index much more than macroeconomic ones. Therefore, we will create an additional model that includes a composite index indicating overall renovations in a time period. We will also include a producer price index as a proxy variable for consumers burdened by living goods. We will also remove from the model GDP and total house loan volume as neither appears to be statistically significant. Our new model is as follows:

$$HPI_t = \beta_0 + \beta_1 EMP_t + \beta_2 CD_t + \beta_3 ln(IRL_t) + \beta_4 REN_t + \beta_5 PPI_t + \epsilon_t$$

The Dubrin-Watson test, which gives a p-value of nearly zero indicates we have autcorrelation of at least type AR(1). The ACF test indicates we should apply serial autocorrelation robust standard error lag for three periods. Likewise, the ADF test statistics encourage us to correct for trends in the data. Our final results are as follows:

Coefficients	**Estimate**	**Std. Error**	**t value**	**Pr(> $\|t\|$)**
β_0	14.60	5.11	2.85	0.0077 **
β_1	-0.125	0.054	-2.32	0.0275 *
β_2	0.00673	0.00164	4.09	0.0003 ***
β_3	29.80	27.95	1.07	0.295
β_4	0.0109	0.00745	1.47	0.153
β_5	0.803	1.475	0.54	0.590
Residual standard error: 12.8 on 30 degrees of freedom				
R^2= 0.538, R^2_{adj}= 0.461				
F-statistic: 6.98 on 5 and 30 DF, p-value: 0.000198				

It appears that there is no improvement in the explanatory power of the model. Furthermore, none of our additional variables are statistically significant. For example, we can explain the insignificance of the PPI by the fact that the Czech Republic has a trade surplus and produces a wealth of tradeable goods, which are not consumed by regular consumers. Likewise, other services or significant components of economic activity are not directly consumed by a typical individual. The statistical insignificance of the renovation variable may make sense if we consider that most renovated dwellings are not sold. If the statistical methodology counts as renovations any renovation activity that needs planning permission, given the state of Czech laws, it would not be surprising if that variable represented even minor renovations.

Section E

Our overall results indicate that there is strong evidence that primarily the construction of housing and the employment rate per 1,000 people have a statistically significant impact on the HPI index. Both models suggest that with an additional 1 person per 1,000 Czechs employed, the HPI will decrease somewhere between 0.11 and 0.125 points. For every 1,000 additional homes built, the HPI index is projected to increase by approximately 7 points. One possible explanation to this result is that due to the regulatory enviroment in the Czech Republic, most new buildings are considered to be luxury dwellings, and as a result, prices for them continue to rise due to a so-called luxury premium charged by builders. Both our models explain somewhat less than half the variation in the HPI index. The significance of the effect of house loan interest rates is inconclusive

Possible improvements, especially with better data and more time, could express unemployment as a percentage, rather than as a proportion of total population. If the number of pensioners increases, the unemployment rate will probably not change even if the number of Czechs employed per 1,000 will change. Another possible improvement is to break down housing by value segments or have a variable representing the permissiveness of local authorities in approving building permits. Modeling the impacts of the Russo-Ukraine war would also be interesting, as theoretical arguments for both a positive and a negative effect on the HPI can be put forward.